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Two-way speaker

Abstract

A two-way speaker includes: a frame having a space for accommodating parts; a yoke partitioning the space of the frame vertically and having a lower surface and a side wall; a first speaker unit disposed below the yoke and having a magnet, a top plate, a voice coil, and a diaphragm; a protector disposed below the first speaker unit and protecting the diaphragm; a second speaker unit disposed above the yoke and having a magnetic circuit and a flexible-printed coil membrane (F-PCM) diaphragm having a coil pattern formed on a polymer film; and a sound emission yoke attached to an upper surface of the second speaker unit.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a two-way speaker.
BACKGROUND
(2) FIG. **1** is a diagram illustrating a two-way speaker according to the related art. In the two-way speaker according to the related art, a cover **10** is coupled to an upper side of a frame **11** formed with an open top and a yoke **12** is coupled to an upper side of the frame **11** inside the cover **10** to form a space portion in which a large speaker unit **30** is installed between a yoke bottom plate **12-1** and a bottom surface of the frame **11**.
(3) In the yoke **12**, a yoke cylinder **12-2** is integrally formed on an outer periphery of the yoke

bottom plate **12-1** formed to have a disk shape, a yoke flange **12-3** is integrally formed at an upper end of the yoke cylinder **12-2**, and the yoke flange **12-3** protrudes slightly upwardly relative to an upper surface of the cover **10**.

(4) A sound emission hole **40** is formed in the upper surface of the cover **10** and the yoke flange **12-3** coupled as described above so that sound (consonants) generated by a large speaker unit **30**, which will be described in detail below, is emitted (output) to the outside of a small speaker **20**, which will be described in detail below, through the sound emission hole **40**.

(5) In the yoke **12**, a first magnet **13** equipped with a first magnetic plate **15** is installed on (fixed to) an upper surface of the yoke bottom plate **12-1** so that a first air gap **26** is formed between an outer periphery of the first magnet **13** and the yoke cylinder **12-2**, and the small speaker unit **20** was installed on the yoke flange **12-3** on an upper side of the first magnet **13**.

(6) In the related art, since the yoke **12** is generally machined through a press process, the yoke flange **12-3** is formed by bending the outer peripheral portion of the yoke cylinder **12-2**. Accordingly, a bent portion between the yoke cylinder **12-2** and the yoke flange **12-3** consequently includes a round-shaped section. FIG. **2** is a diagram illustrating magnetic field flow around the round section of the yoke of the 2-way speaker according to the related art. As can be seen in the drawing, the round section formed by the bending is vulnerable to magnetic flux leakage. In addition, a small diaphragm **21** included in the small speaker unit **20** is fixed to the upper surface of the yoke flange **12-3** by a small diaphragm holder **22** having a small outer periphery. Here, there is a disadvantage in that a separate jig should be used to match the concentricity of the small diaphragm **21** and the voice coil **25**.

(7) In order to solve this problem, the present applicant has proposed a two-way receiver having a yoke assembly having a magnetic field condensation structure in Korea Patent No. 10-2252018.

(8) FIGS. **3** and **4** are diagrams illustrating the two-way speaker disclosed in Korean Patent No. 10-2252018. The two-way speaker according to the related art includes a yoke assembly including a yoke **10***a* and a first ceramic plate **12***a* attached to an upper surface of the yoke **10***a*.

(9) A first speaker unit including a first permanent magnet **31**, a first top plate **32**, a first voice coil **41**, and a first diaphragm **51** is installed above the yoke assembly. In addition, a second speaker unit including a second permanent magnet **33**, a second top plate **34**, a second voice coil **42**, and a second diaphragm **52** is installed below the yoke assembly.

(10) However, speakers having a dynamic structure generally have the advantage in low frequencies because they may use large power due to good magnetic field efficiency, but are disadvantageous in high frequencies the weight of a vibrating system that increases due to the relatively heavy voice coils **41** and **42** attached to the diaphragms **51** and **52**. Although a speaker specialized for high-frequency sound, while adopting the dynamic structure, may be designed but has shortcomings that a response speed is slow because power generated by the magnetic field is indirectly transferred to the diaphragms **51** and **52** through the voice coils **41** and **42**.

(11) Therefore, it is required to develop a two-way speaker including a tweeter speaker which is advantageous for high-frequency sound and has a fast response speed.

SUMMARY

(12) An aspect of the present disclosure provides a two-way speaker employing a flexible-printed coil membrane (F-PCM) structure as a tweeter speaker.

(13) According to an embodiment of the present disclosure, a two-way speaker includes: a frame having a space for accommodating parts; a yoke partitioning the space of the frame vertically and having a lower surface and a side wall; a first speaker unit disposed below the yoke and having a magnet, a top plate, a voice coil, and a diaphragm; a protector disposed below the first speaker unit and protecting the diaphragm; a second speaker unit disposed above the yoke and having a magnetic circuit and a flexible-printed coil membrane (F-PCM) diaphragm having a coil pattern formed on a polymer film; and a sound emission yoke attached to an upper surface of the second speaker unit.

(14) As another example of the embodiment, the frame may have a groove portion, serving as a duct for guiding sound from the first speaker unit toward the sound emission yoke, on an inner circumferential surface of the side wall of the frame.
(15) As another example of the embodiment, the sound emission yoke may include a first sound emission hole for emitting sound transferred from the first speaker unit and a second sound emission hole for emitting sound generated by the second speaker unit.
(16) As another example of the embodiment, the frame may have an atmospheric pressure equalization hole on a side thereof to reduce a difference in atmospheric pressure between upper and lower sides of the diaphragm.
(17) As another example of the embodiment, the magnetic circuit of the second speaker unit may include a lower magnet including a lower center magnet attached to a center of an upper surface of the yoke and a lower side magnet having a ring shape and disposed to be spaced apart from the lower center magnet and an upper magnet including an upper center magnet attached to a center of a lower surface of the sound emission yoke and an upper side magnet having a ring shape and disposed to be spaced apart from the upper center magnet, and the F-PCM diaphragm may be located between the lower magnet and the upper magnet.
(18) As another example of the embodiment, the magnet of the first speaker unit may have a magnetization direction the same as magnetization directions of the lower side magnet and the upper center magnet of the second speaker unit and have a magnetization direction opposite to magnetization directions of the lower center magnet and the upper side magnet.
(19) Since the two-way speaker according to the embodiment employs a speaker including a flexible-printed coil membrane (F-PCM) diaphragm as a tweeter speaker handling high-frequency range of sound, power based on a magnetic field may be directly transferred to the F-PCM that serves as both a diaphragm and a coil and reduces the weight of a vibrating system compared with a dynamic structure, which is advantageous for high-frequency sound reproduction.
(20) In addition, the two-way speaker provided by the embodiment requires additional parts and processes because speakers having different structures are separately manufactured and then combined using additional injection-molded products, but in the embodiment, both the first speaker unit and the second speaker unit are installed on the yoke to form a single unit, thereby simplifying the manufacturing process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a view illustrating a two-way speaker according to a related art;
(2) FIG. **2** is a diagram illustrating a magnetic field flow around a round section of a yoke of the 2-way speaker according to the related art;
(3) FIGS. **3** and **4** are diagrams illustrating a two-way speaker disclosed in another related art;
(4) FIG. **5** is an exploded view of a two-way speaker according to an embodiment;
(5) FIG. **6** is a cross-sectional view of a two-way speaker according to an embodiment;
(6) FIG. **7** is a diagram illustrating a flexible-printed coil membrane (F-PCM) diaphragm provided in a two-way speaker according to an embodiment;
(7) FIG. **8** is a diagram illustrating a frame provided in a two-way speaker according to an embodiment;
(8) FIG. **9** is a perspective view of a two-way speaker according to an embodiment;
(9) FIG. **10** is a view illustrating a two-way speaker installed in an ear set according to an embodiment;
(10) FIG. **11** is a diagram illustrating a magnet polarity arrangement of a two-way speaker according to an embodiment;

(11) FIG. **12** is a diagram illustrating magnetic flux density of a dynamic speaker provided in a two-way speaker according to an embodiment; and

(12) FIG. **13** is a diagram illustrating magnetic flux density when a dynamic speaker provided in a two-way speaker and an F-PCM speaker are installed together according to an embodiment.

DETAILED DESCRIPTION

(13) Hereinafter, embodiments are described in more detail with reference to the drawings.

(14) FIG. **5** is an exploded view of a two-way speaker according to an embodiment, and FIG. **6** is a cross-sectional view of a two-way speaker according to an embodiment.

(15) A two-way speaker according to an embodiment includes a first speaker unit employing a dynamic speaker and a second speaker unit employing a flexible-printed coil membrane (F-PCM) vibrating system. The first speaker unit and the second speaker unit are installed in one frame **100** and share a yoke **210**.

(16) The first speaker unit employs a dynamic speaker structure of the related art, and a magnetic circuit including the yoke **210**, a magnet **220**, and a top plate **230** is installed in the frame **100**. The yoke **210** has a circular top surface **212** and a cylindrical side wall **214**, and the side wall **214** is coupled to an inner circumferential surface of the side wall **110** of the frame **100**.

(17) A magnetic gap is formed between the sidewall **214** of the yoke **210** and the magnet **220**, and an upper end of the voice coil **300** is located in the magnetic gap. The upper end of the voice coil **300** is attached to a diaphragm **400**. The frame **100** is provided with a seating end **120** on which an outer periphery of the diaphragm **400** is seated. A voice coil **300** is attached to the diaphragm **400**, and when an electric signal is applied to the voice coil **300**, the voice coil vibrates by mutual electromagnetic force with a magnetic circuit. Here, the diaphragm **400** to which the voice coil **300** is attached also vibrates and generates sound. In an embodiment, the first speaker unit may be one-magnet type speaker unit in which a magnetic gap is formed between the sidewall **214** of the yoke **210** and the magnet **220**, or a two-magnet type, a three-magnet type, or a five-magnet type dynamic speaker may be employed as needed.

(18) At the lowermost side, a protector **500** is coupled to the frame **100** to surround the diaphragm **400** of the first speaker unit. In detail, the protector **500** surrounds the outer periphery of the frame **100** and is attached to an upper surface of the diaphragm **400** to protect the first speaker unit.

(19) Meanwhile, an F-PCM type second speaker unit is installed above the yoke **210**. The second speaker unit includes a magnetic circuit and an F-PCM **700**.

(20) The second speaker unit includes the yoke **210**, magnets **610**, **620**, **630**, and **640**, and a sound emission yoke **800**, as a magnetic circuit.

(21) The magnets **610**, **620**, **630**, and **640** include lower magnets **610** and **620** including a lower center magnet **610** attached to the center of an upper surface of the yoke **210** and a lower side magnet **620** having a ring shape and disposed to be spaced apart from the lower center magnet **610** and upper magnets **630** and **640** including an upper center magnet **630** attached to the center of a lower surface of the sound emission yoke **800** and an upper side magnet **640** having a ring shape and disposed to be spaced apart from the upper center magnet **630**. The F-PCM **700** is disposed between the lower magnets **610** and **620** and the upper magnets **630** and **640**.

(22) FIG. **7** is a diagram illustrating an F-PCM diaphragm included in a two-way speaker according to an embodiment.

(23) The F-PCM **700** has a conductive pattern **720** formed on a polymer film **710**, and serves as both a diaphragm and a voice coil in a dynamic speaker. The F-PCM **700** is very thin compared to the total height of the voice coil, so it is advantageous for miniaturization, and since the F-PCB **700** is light in weight to be advantageous for high-frequency sound reproduction, the F-PCM **700** may be advantageously used as a tweeter speaker.

(24) Meanwhile, instead of having a separate terminal, the F-PCM **700** may be elongated to form a terminal portion **730**, and the terminal portion **730** may be connected to an external power source or an external circuit or may be connected to a control circuit of a two-way speaker. As described

above, the sound emission yoke **800** is attached to the upper side of the lower magnets **610** and **620**. In this case, the sound emission yoke **800** may have a sound emission structure so that not only sound generated by the second speaker unit but also sound generated by the first speaker unit may be emitted.

(25) FIG. **8** is a view illustrating a frame provided in a two-way speaker according to an embodiment, FIG. **9** is a perspective view of a two-way speaker according to an embodiment, and FIG. **10** is a view illustrating a two-way speaker installed in an ear set according to an embodiment.

(26) The frame **100** includes a side wall **110** having an inner circumferential surface coupled to an outer surface of the yoke **210** and outer surfaces of the lower side magnet **620** and the upper side magnet **640**. In this case, one or more groove portions **130** serving as a duct for guiding sound reproduced by the first speaker unit to the sound emission yoke **800** are formed on the inner circumferential surface of the side wall **110**. The groove portion **130** is connected to a first sound emission hole **810** formed in the sound emission yoke **800**, so that sound reproduced by the first speaker unit is emitted to the first sound emission hole **810**.

(27) In addition, sound reproduced by the second speaker unit is emitted through a second sound emission hole **820** formed in a position corresponding to a gap between the upper center magnet **630** and the upper side magnet **640**. That is, the sound reproduced by the first speaker unit and the sound reproduced by the second speaker unit are emitted in the same direction.

(28) Meanwhile, when the two-way speaker is installed in an ear set, a diaphragm **400** of the first speaker unit, which is a dynamic speaker, vibrates, and a difference in atmospheric pressure between an ear canal and the outside may cause deafness or pain in the ear. In order to prevent this, the frame **100** includes an atmospheric pressure equalization hole **140** penetrating through the side wall **110**. The atmospheric pressure equalization hole **140** is formed in a position that may communicate with an upper space of the diaphragm **400** of the first speaker unit. At this time, a mesh or a filter may be installed in the atmospheric pressure equalization hole **140** to adjust air permeability of air flowing in and out through the atmospheric pressure equalization hole **140**.

(29) FIG. **10** is a view illustrating a two-way speaker installed in an earphone according to an embodiment, in which the red arrow indicates a sound emission path of sound generated by the first speaker unit, the blue arrow indicates a sound emission path of sound generated by the second speaker unit, and the black arrow indicates an air flow path for atmospheric pressure equalization.

(30) FIG. **11** is a diagram illustrating a magnet polarity arrangement of a two-way speaker according to an embodiment, FIG. **12** is a diagram illustrating a magnetic flux density of a dynamic speaker provided in a two-way speaker according to an embodiment, and FIG. **13** is a diagram illustrating magnetic flux density when a dynamic speaker provided in a two-way speaker and an F-PCM speaker are installed together according to an embodiment.

(31) As described above, each speaker unit of the two-way speaker shares the yoke **210**. Accordingly, the number of required parts may be reduced and a product assembly process may be simplified. In addition, the efficiency of a magnetic field shared by the two speakers may be increased by controlling a magnetization direction of the magnet **220** of the first speaker unit and the magnets **610**, **620**, **630**, and **640** of the second speaker unit. For example, as shown in FIG. **11**, a magnetic flux density may be improved by controlling a magnetization direction of the magnet **220** of the first speaker unit to be the same as those of the lower side magnet **620** and the upper center magnet **630** of the second speaker unit and to be the opposite to those of the lower center magnet **610** and the upper side magnet **640** of the second speaker.

(32) It can be seen that an average magnetic flux density B is increased by about 10% when the dynamic speaker and an F-PCM speaker sharing a magnetic field with the dynamic speaker are installed together, compared to a case in which only the dynamic speaker is installed as shown in FIG. **12**. When the magnetic flux density increases, a magnetic force may increase and power of the speaker may be improved consequently.

(33) As used herein, the terms “having,” “containing,” “including,” “comprising,” and the like are

open-ended terms that indicate the presence of stated elements or features, but do not preclude additional elements or features. The articles “a,” “an” and “the” are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.

(34) It is to be understood that the features of the various embodiments described herein may be combined with each other, unless specifically noted otherwise.

(35) Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

Claims

1. A two-way speaker, comprising: a frame having a space for accommodating parts; a yoke partitioning the space of the frame vertically and having a lower surface and a side wall; a first speaker unit disposed below the yoke and having a magnet, a top plate, a voice coil, and a diaphragm; a protector disposed below the first speaker unit and protecting the diaphragm; a second speaker unit disposed above the yoke and having a magnetic circuit and a flexible-printed coil membrane (F-PCM) diaphragm having a coil pattern formed on a polymer film; and a sound emission yoke attached to an upper surface of the second speaker unit.
2. The two-way speaker of claim 1, wherein the frame has a groove portion that forms a duct for guiding sound from the first speaker unit toward the sound emission yoke, on an inner circumferential surface of the side wall of the frame.
3. The two-way speaker of claim 2, wherein the sound emission yoke includes a first sound emission hole for emitting sound transferred from the first speaker unit and a second sound emission hole for emitting sound generated by the second speaker unit.
4. The two-way speaker of claim 1, wherein the frame has an atmospheric pressure equalization hole on a side of the frame to reduce a difference in atmospheric pressure between upper and lower sides of the diaphragm.
5. The two-way speaker of claim 1, wherein the magnetic circuit of the second speaker unit comprises: a lower magnet including a lower center magnet attached to a center of an upper surface of the yoke and a lower side magnet having a ring shape and disposed to be spaced apart from the lower center magnet; and an upper magnet including an upper center magnet attached to a center of a lower surface of the sound emission yoke and an upper side magnet having a ring shape and disposed to be spaced apart from the upper center magnet, and wherein the F-PCM diaphragm is located between the lower magnet and the upper magnet.
6. The two-way speaker of claim 5, wherein the magnet of the first speaker unit has a magnetization direction which is the same as magnetization directions of the lower side magnet and the upper center magnet of the second speaker unit, and has a magnetization direction opposite to magnetization directions of the lower center magnet and the upper side magnet.